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A multifunctional geometric phantom for digital radiography assessment

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ABSTRACT

Introduction: Geometric accuracy in digital radiography is essential for maintaining image quality, diagnostic reliability, and regulatory compliance. Misalignments in beam orientation can introduce distortion, affect spatial measurements, and compromise diagnostic outcomes. Despite its importance, current quality control (QC) tools rarely provide integrated assessments of all relevant geometric parameters.

Methods: A modular three-layer geometric phantom was developed to assess beam perpendicular, beam area alignment, magnification, and object displacement. Radiographic tests were conducted using varied focus-to-detector distance (*FDD*) (50, 60, and 70 cm). Measurements were extracted from radiographic images using image processing software, and statistical analyses were applied to evaluate the relationship between magnification and displacement.

Results: At shorter *FDD*, beam deviation increased, with a maximum angular deviation of 0.28° at 50 cm. Beam alignment deviation ranged from 2.00 % at 50 cm to 1.29 % at 70 cm, all within acceptable limits. Object magnification increased significantly with greater object-detector distance and smaller *FDD*. A strong linear correlation was found between magnification and displacement ($r = 0.9987$, $p < 0.0001$), with an estimated displacement increase of ± 0.56 mm for every 0.01 magnification unit.

Conclusion: The proposed phantom effectively quantifies geometric deviations in digital radiography, producing consistent and statistically robust measurements across varied imaging conditions. These findings highlight its potential as a reliable and practical tool for quality control applications.

Implication for practice: This phantom offers a cost-effective, adaptable solution for routine QC procedures in radiographic imaging. Its ability to directly quantify geometric errors supports more precise calibration, reduces the risk of image distortion, and enhances diagnostic confidence. Adoption of this tool in clinical and regulatory settings can improve patient safety and standardize imaging performance evaluation.

Introduction

Radiography is a critical diagnostic technique in the medical field, enabling the visualization of the internal contents of objects using X-rays.[1,2] Generally, there are several types of radiography, including conventional, digital, fluoroscopy, and panoramic.[3,4] Digital radiography is a variant of conventional radiography, where the resulting images are in a digital format, allowing for rapid and efficient image acquisition.[5–7] A digital radiographic image is considered high-quality if it meets specific visibility and geometric parameters.[8,9] The visibility quality parameters of a digital image include brightness, contrast, noise, signal-to-noise ratio (SNR), and artefacts.[10] Conversely, the geometric parameters for assessing digital image quality encompass sharpness, magnification, and shape distortion.[11] To maintain consistent image quality from a digital radiography device, regular quality control (QC) is essential.[12,13]

QC guidelines for digital radiography have been developed by organizations such as the International Atomic Energy Agency (IAEA), the American Association of Physicists in Medicine (AAPM), and the European Federation of Organizations for Medical Physics (EFOMP).[12,14,15] These protocols emphasize periodic

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assessments of detector performance, radiation dose, and geometric accuracy to uphold diagnostic reliability and patient safety.[16] In Indonesia, the Decree of the Minister of Health No. 1250/MENKES/SK/XII/2009 mandates regular conformity testing of radiographic systems. This includes evaluations of the collimator, X-ray tube, generator, and automatic exposure control.[17] Among these components, collimator testing plays a crucial role. Errors in collimation can directly impact image accuracy and field alignment, resulting in diagnostic uncertainty or increased radiation exposure.[18,19] However, the testing of X-ray systems, particularly collimator evaluation, predominantly employs conventional methods. Assessments utilize standard regulatory tools, such as the Collimator and Alignment Test tools.[12]

These testing instruments face significant challenges related to their application. A notable limitation of these tools is their susceptibility to measurement inaccuracies, which arise mainly from their limited resolution and the effects of beam divergence and scatter that blur field boundaries.[20,21] The main contributing factors to these inaccuracies are the inability of the tools to resolve fine measurement scales and their lack of compensation for beam divergence and scattered radiation. Another important limitation is that the current tools cannot provide an exact quantification of light–X-ray field congruence or central beam angular deviation; instead, they only rely on reference markings. For example, in beam alignment testing, the deviation is considered acceptable if the point remains within the reference circle, which corresponds to an angular deviation of less than 3°.[22] Research indicates that collimation tests using these tools often yield considerable errors.[23] Additionally, the measurement capabilities of the collimator and alignment test tool are quite limited. Currently, these tools can only identify the magnitude of the collimator's angular deviation and the extent of X-ray displacement.[12,24] Moreover, current tools do not directly assess geometric parameters such as object magnification and displacement, which are critical for diagnostic precision and dose optimization.[23]

Although geometric magnification is a well-known and theoretically predictable phenomenon in projection X-ray imaging, its practical quantification in clinical settings and routine QC remains limited.[25] Theoretically, magnification can be estimated using the object-to-image distance (OID) and source-to-image distance (SID), assuming all geometric components are known and fixed.[26–29] However, in practice, the precise positions of the X-ray focus, object, and detector plane are often not readily available or reliably maintained, especially in systems without integrated spatial calibration. Furthermore, the assumption that magnification behaves uniformly across object sizes and volumes fails in many clinical scenarios, where larger or three-dimensional objects exhibit non-uniform magnification, making it difficult to estimate true scale or geometry from image data alone.[30,31] Accurate geometric interpretation is critical for dose estimation and therapeutic decision-making in clinical scenarios, particularly in orthopedic imaging, pediatric radiology, and oncologic planning.[32–34]

Although AAPM Task Group 150 addresses QC procedures for digital radiography, it does not prescribe any specific phantom design capable of directly quantifying spatial geometry or magnification factors.[35] Instead, current frameworks assume known system geometry or rely on indirect assessment methods, which are insufficient for detecting misalignments or spatial distortions across volumetric structures. Most phantom-based QC studies have focused on contrast resolution, noise, and detector uniformity, leaving a gap in standardized tools for geometric validation.[36–38]

To address this gap, this study introduces a multifunctional geometric phantom designed for conformity testing in digital radiography systems, with particular emphasis on collimator performance and geometric accuracy. Unlike traditional tools that are flat, single-layered, and restricted to a single test point, thereby preventing evaluation of magnification and displacement, the proposed phantom features a three-layer modular cube design, enabling controlled variation of object-to-detector distance. This multifunctionality allows simultaneous evaluation of X-ray beam perpendicular, beam alignment, magnification factor, and geometric displacement, overcoming the inherent limitations of conventional test tools. In addition, the phantom is highly flexible because its modular layers can be assembled or disassembled as needed, allowing tailored configurations for different testing requirements.

In contrast, the proposed phantom bridges this gap by providing absolute, geometry-based magnification measurements that can be scaled or adapted to more complex clinical objects. This allows for more robust system validation, including potential applications in software calibration, geometric correction algorithms, and even dose modelling. This study does not aim to validate a radiography system or compare results with standard commercial tools. Instead, the focus lies on developing and initially evaluating a phantom-based QC method using theoretical geometric principles as the reference. This approach represents an early-stage validation demonstrating the tool's internal consistency and measurement feasibility. However, to ensure clarity and analytical focus, this paper specifically addresses the evaluation of beam perpendicular, beam alignment, magnification, and object displacement; broader applications such as contrast, penumbra, and spatial resolution will be explored in future studies.

Methods

Design and fabrication geometric phantom

In this study, we developed a geometrical phantom from acrylic material ($\rho = 1.18$ g/cm^3),[39] with a thickness of 100 mm as a test object, as shown in Fig. 1 (b). The phantom consists of three-layered sections, top, middle, and bottom, with 105 mm between the layers. Each side of the phantom layer measures 21 cm in length and width, while the height of the phantom is also 21 cm. Each layer, top, middle, and bottom, features nine holes with a diameter of 5 mm, arranged in a square pattern with a spacing of 5 cm between the holes. An aluminium cylinder with a diameter of 5 mm and a height of 6 mm is inserted into each hole as the test guide object. Aluminium was selected as the test object material because it provides shades of grey comparable to clinical radiographs and is widely recognized as the standard for step-wedge phantoms in contrast testing.[35,40,41] While other materials such as acrylic, PMMA, or copper were considered, aluminium offers suitable attenuation within the diagnostic energy range along with practical advantages of availability, ease of fabrication, cost-effectiveness, and durability for repeated use.[42,43]

Experimental setup

The phantom was tested using a digital radiography system (HF100B-MDN/5012A, PT. Madeena Karya Indonesia) with a fluorescence detector and a 4096×3296-pixel resolution camera. The beam perpendicular and beam area alignment tests were conducted using three focus-to-detector distance variations (*FDD*): 70 cm, 60 cm, and 50 cm, with an exposure setting of 70 kV and 8 mAs. Although 100 cm is widely used in standard protocols and quality control testing,[35] radiographers in clinical practice frequently adopt shorter distances when performing patient examinations, particularly in situations requiring flexibility or when space is constrained. Therefore, these *FDD* values were chosen to

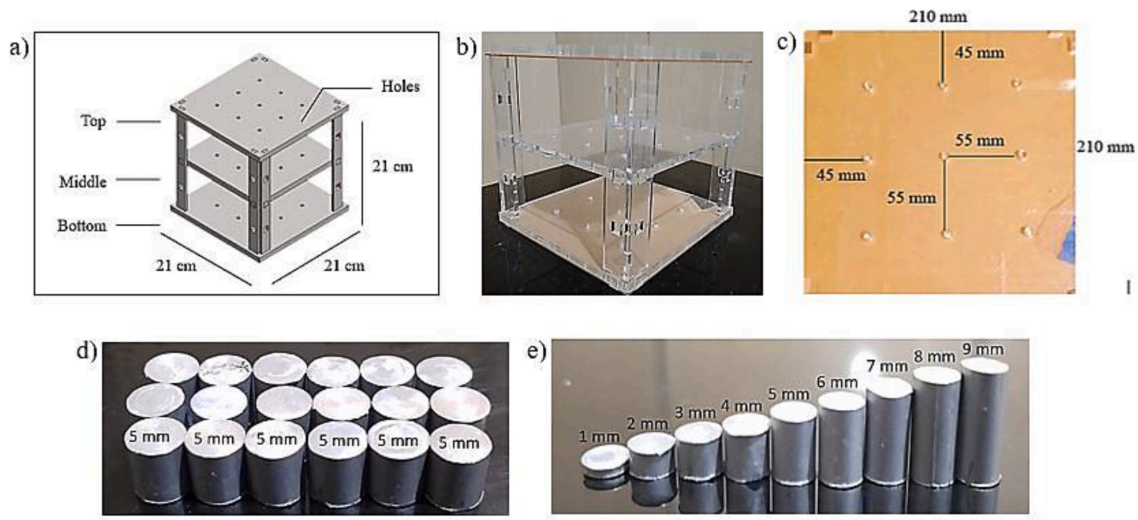

Figure 1. The test object used in this study, consisting of: (a) the schematic design of the phantom, (b) the constructed geometric phantom, (c) the layered view of the phantom, (d) the aluminum cylindrical test object utilized in the study, and (e) aluminum cylinders of varying thicknesses used for contrast testing, all of which were inserted into the corresponding holes in the phantom layers.

Figure 2. Phantom placement position prior to the scanning process. The phantom must be positioned perpendicular to the X-ray source to prevent geometric distortion.

represent realistic clinical usage conditions. In the beam perpendicular test, the phantom was positioned perpendicular to the X-ray source (Fig. 2), and an inclinometer ensured a level setup. The collimator light aligned the shadow edges with the test object, and the irradiation field was set to 21 × 21 cm (Fig. 3). Irradiation was performed once at each *FDD* to observe the influence of distance on beam deviation angles. Although conventional testing typically uses a single distance of 100 cm, in practical applications, shorter distances such as 50 cm, 60 cm, and 70 cm are also commonly used. The beam area alignment test employed the same setup and *FDD* variations. However, the irradiation field was reduced to 11 × 11 cm (Fig. 3) to align precisely with the phantom hole edges, allowing for accurate measurement of beam alignment deviations. After proper configuration, irradiation was conducted for each *FDD* variation. In this study irradiation was carried out by a certified radiographer from PT. Madeena Karya Indonesia, the provider of the digital radiography system. Subsequent image analysis and parameter measurements were performed by the research team.

Image processing

All radiographic images were processed using ImageJ v1.54, focusing on flat field correction (FFC) and binary segmentation.

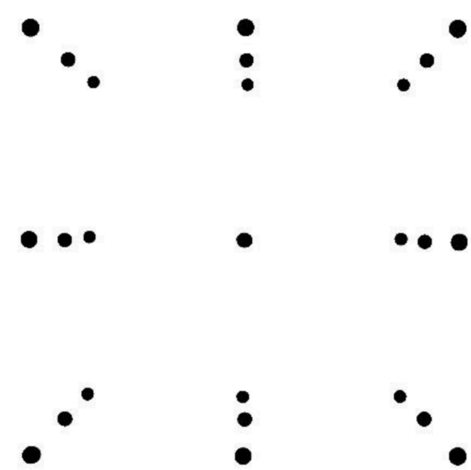

Figure 3. Binary segmentation image for magnification and displacement analysis.

FFC was performed by acquiring a dark image (no radiation) and a gain image (uniform radiation without object), to minimize detector noise and intensity inhomogeneity. The corrected images were converted to 8-bit grayscale and subjected to global thresholding, where the optimal threshold value was selected empirically and adjusted for varying exposure levels at different source-to-detector distances. Pixels above the threshold were classified as foreground (white), and others as background (black), producing a binary image suitable for magnification and displacement analysis (Fig. 3).

Calibration factor

Before measurement, a calibration factor was established using TOR phantom images (Fig. 4), where each box represents 0.5 cm. This yielded a calibration of 100 pixels per centimeter (10 pixels per millimeter), enabling accurate conversion from pixels to real-world units. This calibration step was essential to ensure the dimensional accuracy of subsequent image analyses, particularly for calculating object diameter, magnification, and displacement.

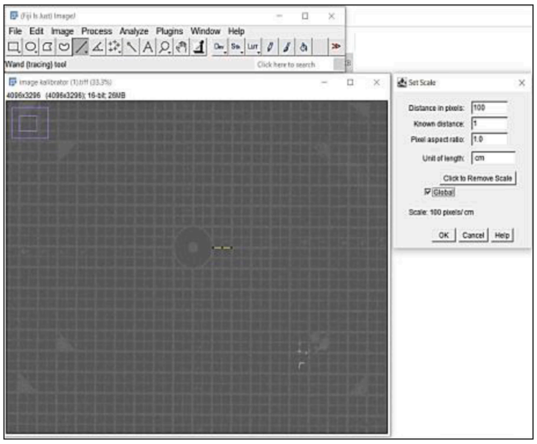

Figure 4. An image of the TOR phantom is used as the calibration factor in this study.

Measurement

This study focused on four geometric analyses: beam perpendicular, beam area alignment, test point magnification, and displacement. All measurements required for these geometric calculations were performed using ImageJ v1.54, an open-source image analysis software widely validated in radiological and biomedical research.

Beam perpendicular to the central point

To assess the central ray perpendicular, grayscale radiographic images were acquired at three *FDD* distances using a phantom containing concentric alignment markers. The geometric analysis was performed by projecting the displacement of the beam center using a trigonometric model. Specifically, the angular deviation (θ) of the X-ray central axis was determined by measuring the shift of the projected center point from the ideal central line on the image plane. This deviation was quantified as (Eq. (2)), with a maximum permissible deviation is < 3.[35,44]

$$\theta = arc\ tan^{-1}\left(\frac{r_{deviation}\ x\ SOD}{h_{phantom}\ x\ FDD}\right) \tag{1}$$

Beam Area Alignment

Beam Area Alignment is a test used to assess the congruence between the X-ray field and the collimator light field projected onto the imaging detector. In this study, the test was conducted by acquiring radiographic images of a phantom with visible aluminium markers positioned near the edges of a collimated area. The misalignment was calculated using the formula (Eq. 2) for horizontal axes and (Eq. (3)) for vertical axes. These deviations were then expressed as a percentage of the Source to Image Distance (SID) and compared against the maximum permissible limit of 3 % of SID.[35,44]

$$\Delta X_n = \frac{|X_{n\ collimation} - X_{n\ X-ray}|}{SID}\ x\ 100\ \% \tag{2}$$

$$\Delta Y_n = \frac{|Y_{n\ collimation} - Y_{n\ X-ray}|}{SID}\ x\ 100\ \% \tag{3}$$

Test point magnification and displacement

In this study, the test was conducted using a phantom embedded with aluminium discs of known diameter (5 mm) placed within the collimated field. To streamline the analysis, this study introduced a semi-automated method using a custom ImageJ macro, allowing for faster and reproducible magnification and displacement values across multiple datasets. The magnification factor M was calculated using the formula (Eq. (4)), and the displacement $\Delta D(x,y)$ was calculated using the formula (Eq. (5)). The results were then evaluated against the regulatory tolerance of ±2 % for magnification.[44]

$$M = \frac{Object\ diameter\ on\ image}{Actual\ diameter} \tag{4}$$

$$\Delta D(x,y) = Object\ distance\ on\ image - Actual\ distance \tag{5}$$

Statistical analysis

Statistical analysis was conducted exclusively for the displacement test to determine the relationship between magnification and displacement. The other three tests, beam perpendicular, beam area alignment, and magnification, were assessed using direct geometric measurements based on established equations. For each *FDD* setting, irradiation was performed once, and the subsequent measurement of test parameters was repeated five times to ensure reliability and minimize operator-dependent variability. All data in this study were presented as mean ± SD values. Linear associations between magnification (times) and test-point displacement (mm) were quantified with Pearson correlation and simple ordinary-least-squares (OLS) regression.[45,46] Residual normality was evaluated using Q–Q plots and the Shapiro–Wilk test; homoscedasticity with the Breusch–Pagan test.[47]

Results

Analysis of beam perpendicular to the central point

Fig. 5 illustrates the test results across focus-to-detector distances (*FDD*). The analysis highlights a shift in the X-ray beam's central point, as indicated by the displacement of the central circle within the images. This deviation underscores the phantom's capacity to detect beam misalignment and direction with high sensitivity, a level of precision beyond the capability of conventional tools.

Table 1 summarizes the angular deviation of the X-ray beam's central point at *FDD* of 50 cm, 60 cm, and 70 cm, detailing object height, deviation distance, angle, and verification status based on the 3° acceptance threshold. All values remain within this limit, affirming the method's effectiveness in validating system performance.

Analysis of beam area alignment

In the second test, we evaluated the conformity of beam area alignment. Fig. 6 shows the radiographic images from the irradiation field test. Visually, all three images indicate a shift in the position of the test object relative to the irradiation field. Each *FDD* distance results in a varying shift characteristic.

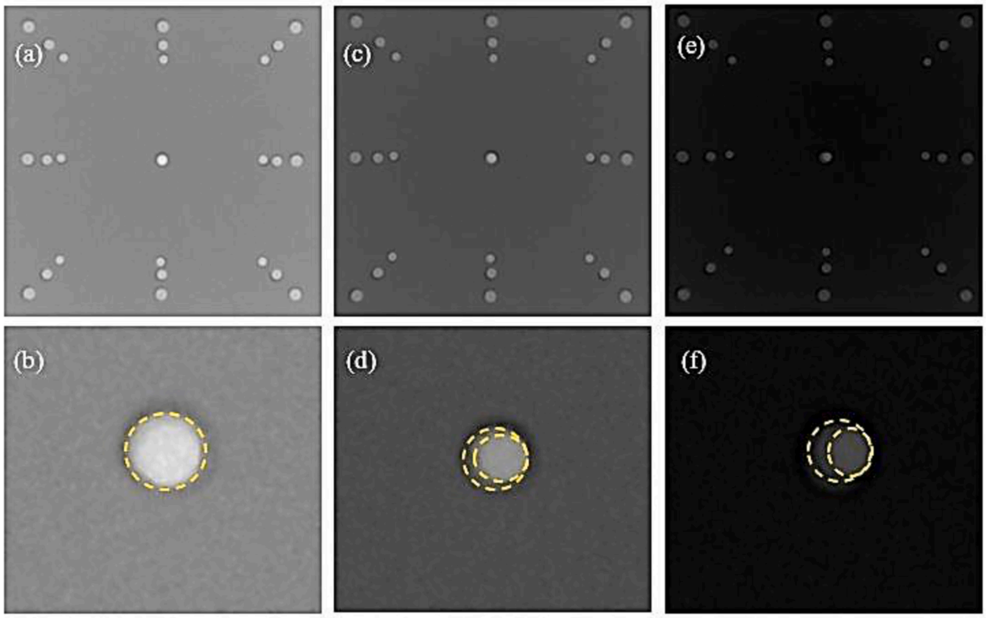

Figure 5. The test results on the geometric phantom show that reducing the *FDD* causes a shift in the X-ray beam's central point. Specifically: (a) shows the radiographic image at 70 cm *FDD*, (b) the central point image at 70 cm, (c) the radiographic image at 60 cm, (d) the central point image at 60 cm, (e) the radiographic image at 50 cm, and (f) the central point image at 50 cm. This shift confirms that the central point deviation increases as *FDD* decreases.

Table 1
Results of perpendicular testing for the central point of the X-Ray beam using *FDD* variations.

No.	*FDD* Distance (cm)	Object Height (cm)	Deviation Distance (mm)	Deviation Angle (°)	Verification
1	70	21	0.00 ± 0.01	0.00	<3⁰ (Accepted)
2	60	21	0.39 ± 0.01	0.07	<3⁰ (Accepted)
3	50	21	1.79 ± 0.01	0.28	<3⁰ (Accepted)

Beam area alignment tests were conducted at *FDD* of 70 cm, 60 cm, and 50 cm, as detailed in Tables 2–4. The tests measured the alignment between the collimator light field and the X-ray radiation field, expressed as displacement relative to the source to image distance (%SID). All values remained within regulatory limits, confirming that the X-ray collimation system meets alignment standards.

Analysis of test point magnification

Table 5 summarizes the magnification analysis of test points across different phantom layers under varying *FDD* conditions. The results in Table 5 indicate that variations in *FDD* and the distance of test points from the detector, represented by the phantom layers, significantly affect the magnification factor.

Fig. 7 illustrates the relationship between *FDD*, test point distance, and magnification percentage. Fig. 7(a) shows that the magnification percentage decreases as *FDD* increases. At 50 cm *FDD*, the magnification percentage was 6.80 % in the lower layer and 50.55 % in the upper layer. Increasing *FDD* to 60 cm reduced these values to 2.59 % and 40.92 %, respectively, while at 70 cm *FDD*, magnification further declined to 0.85 % in the lower layer and 33.07 % in the upper layer. Fig. 7(b) shows the relationship between the test points' distance from the detector and magnification. As the distance increases, magnification rises. At an *FDD* of 70 cm, the magnification is 0.85 % at 0 mm, 13.07 % at 105 mm, and 33.07 % at 210 mm from the detector. At an *FDD* of 50 cm,

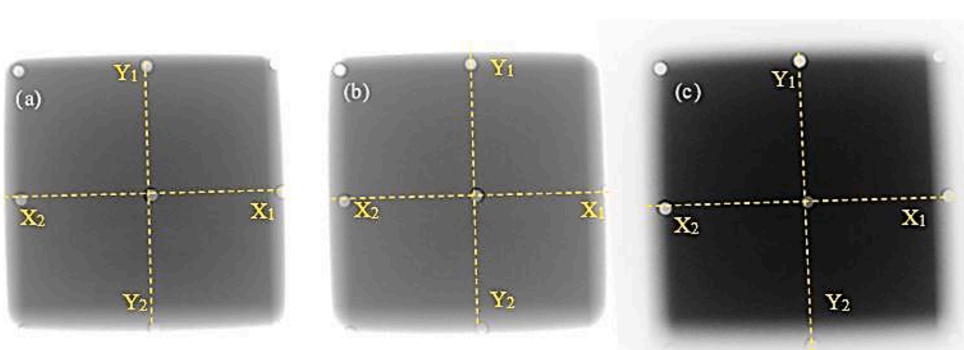

Figure 6. Radiographic images of beam area alignment testing: X1 represents the distance from the right edge to the centre point, X2 represents the distance from the left edge to the centre point, Y1 represents the distance from the top edge to the centre point, and Y2 represents the distance from the bottom edge to the centre point. (a) Image at an *FDD* of 70 cm, (b) at an *FDD* of 60 cm, and (c) at an *FDD* of 50 cm.

Table 2
Results of beam area alignment testing at an *FDD* of 70 cm.

Field Edges	Edge of Light (mm)	Edge of X-Ray (mm)	Magnitude of Shift (%SID)	Δ1+Δ2 (%SID)	X + Y (%SID)	Verification
X_1	55	51.0 ± 0.1	0.56	0.86	1.29	<3 % Accepted
X_2	55	57.0 ± 0.1	0.29			
Y_1	55	56.0 ± 0.1	0.14	0.43		
Y_2	55	53.0 ± 0.1	0.28			

Table 3
Results of beam area alignment testing at an *FDD* of 60 cm.

Field Edges	Edge of Light (mm)	Edge of X-Ray (mm)	Magnitude of Shift (%SID)	Δ1+Δ2 (%SID)	X + Y (%SID)	Verification
X_1	55	49.0 ± 0.1	0.99	1.49	1.66	<3 % Accepted
X_2	55	58.0 ± 0.1	0.50			
Y_1	55	56.0 ± 0.1	0.17	0.17		
Y_2	55	55.0 ± 0.1	0.00			

Table 4
Results of beam area alignment testing at an *FDD* of 50 cm.

Field Edges	Edge of Light (mm)	Edge of X-Ray (mm)	Magnitude of Shift (%SID)	Δ1+Δ2 (%SID)	X + Y (%SID)	Verification
X_1	55	52.0 ± 0.1	0.60	1.21	2.00	<3 % Accepted
X_2	55	58.1 ± 0.1	0.61			
Y_1	55	58.0 ± 0.1	0.60	0.79		
Y_2	55	54.0 ± 0.1	0.19			

magnification increases from 6.80 % to 23.77 % and 50.55 %, respectively. Fig. 5(a) and (b) illustrate how *FDD* and test point distance influence magnification.

Analysis of test point displacement

In this study, the displacement along the *X* and *Y* axes was analyzed, assuming uniform X-ray spread, as shown in Fig. 8. The average displacement along both axes was then calculated to determine the overall displacement for each *FDD* variation.

Table 6 summarizes the average displacement of test points relative to the central point. Displacement was minimal in the lower layers and increased substantially in the middle and upper layers, reflecting greater distances from the detector. In the lower layer, displacement remained close to the actual distance, while in the middle and upper layers, displacement increased notably as *FDD* decreased.

In this study, we analyze the correlation between magnification and displacement. Fig. 9 presents the statistical analysis; a significant linear association was observed between magnification and test-point displacement. Pearson correlation analysis yielded a coefficient of $r = 0.9987$ with a p-value $= 2.37 \times 10^{-10}$. The model demonstrated an excellent fit, with a coefficient of determination of $R^2 = 0.997$, suggesting that magnification accounts for 99.7 % of the variance in displacement. Both the intercept and slope coefficients were statistically significant ($p < 0.0001$). The Shapiro–Wilk test showed no evidence of residual non-normality ($W = 0.8800$, $p = 0.1570$), and this was visually confirmed by the Q–Q plot of the residuals (Fig. 9 (b)). Additionally, the Breusch–Pagan test indicated homoscedasticity of the residuals ($BP = 0.0174$, $p = 0.895$), supported by the residuals vs. fitted values plot (Fig. 9 (c)), which showed no visible pattern.

Discussion

This study comprehensively evaluates four fundamental geometric parameters in digital radiography: beam perpendicular,

Table 5
The diameter and average magnification of test points on the radiographic image of the geometric phantom.

No.	Layers	Original Diameter (mm)	Diameter of the Image (mm)			Magnification of Test Points (times)		
			FDD 70 cm	*FDD* 60 cm	*FDD* 50 cm	*FDD* 70 cm	*FDD* 60 cm	*FDD* 50 cm
1	Bottom	5.0	5.0 ± 0.1	5.1 ± 0.1	5.3 ± 0.1	1.01 ± 0.01	1.03 ± 0.01	1.07 ± 0.01
2	Middle	5.0	5.7 ± 0.1	5.9 ± 0.1	6.2 ± 0.1	1.14 ± 0.01	1.18 ± 0.01	1.24 ± 0.01
3	Top	5.0	6.7 ± 0.1	7.0 ± 0.1	7.5 ± 0.1	1.34 ± 0.01	1.41 ± 0.01	1.51 ± 0.01

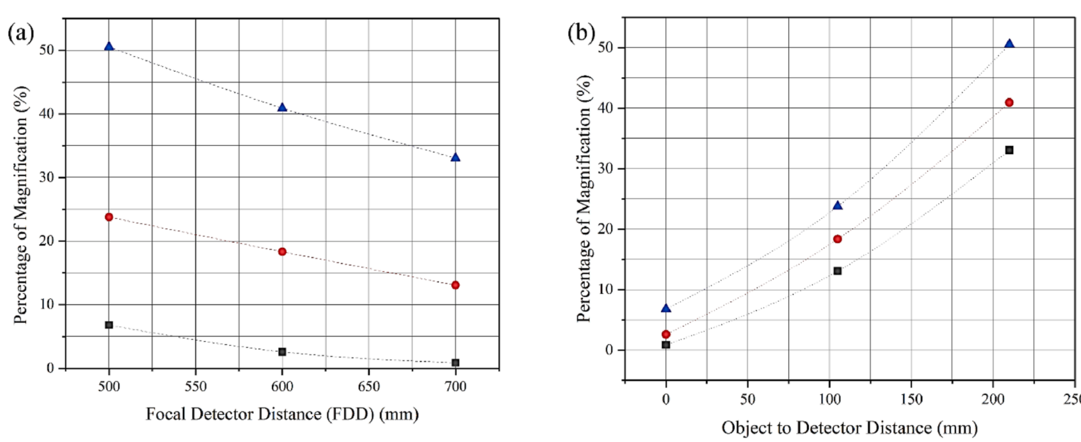

Figure 7. (a) Graph of the relationship between *FDD* and the percentage magnification of the test object in the geometric phantom. The black dots represent the location of the test point in the bottom layer, the red dots indicate the middle layer, and the blue dots represent the top layer. (b) Graph the relationship between the distance from the test point to the detector and the percentage magnification of the test object in the geometric phantom. The black dots indicate an *FDD* of 70 cm, the red dots represent an *FDD* of 60 cm, and the blue dots indicate an *FDD* of 50 cm. Both variables demonstrate a clear relationship with the resulting magnification percentage.

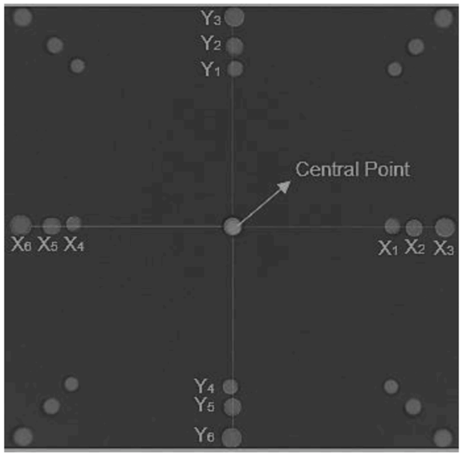

Figure 8. Illustration of the displacement of test points relative to the central point of the geometric phantom due to the distance between the test points and the detector in the digital radiography system. Test points located at the center and upper sections of the phantom tend to exhibit significant positional shifts compared to test points positioned in the lower layers of the phantom.

beam area alignment, magnification, and object displacement. It uses a geometric phantom to simulate variable imaging conditions and evaluate system accuracy. Integrating these parameters is essential to understanding how beam geometry and setup configuration affect image fidelity, diagnostic accuracy, and regulatory compliance. To ensure accurate beam positioning, tests were performed to evaluate the perpendicular of the X-ray beam relative to the detector's central point. Our results confirmed that focus-to-detector distance (*FDD*) directly affects the beam perpendicular. A shorter *FDD* resulted in greater angular deviation, with the highest recorded deviation of 0.28° at 50 cm and no observable deviation at 70 cm. Visual evidence in Fig. 5 further supports these findings, where circular shadow artifacts appear at shorter *FDD*, indicating beam misalignment, while at 70 cm no such deviation is evident. This is because, at shorter distances, the X-ray beam diverges more widely and interacts with objects at more oblique angles, amplifying geometric distortion.[48–50]

This finding is consistent with the study of Brookfield et al. (2015), who reported centering errors across different object-to-detector distances (*ODD*) when evaluating beam alignment with a collimation test tool.[51] Their study identified measurable displacements of the beam center in both the lateral and superior–inferior planes, which varied depending on *ODD*. Taken together, these findings emphasize that geometric misalignments are directly influenced by variations in the spatial relationship between the X-ray source, object, and detector. Both our investigation of *FDD* and the earlier *ODD*-based study highlight that even small deviations in beam centering can amplify geometric distortion and compromise image accuracy, underscoring the necessity of systematic evaluation of beam geometry to ensure diagnostic fidelity and radiation safety.

In parallel, the beam area alignment test assessed the congruence between the X-ray radiation field and the collimator's light field. Our results demonstrated that alignment deviation decreases as *FDD* increases, with a maximum deviation of 2.00 % at 50 cm and the lowest at 1.29 % at 70 cm, all within regulatory limits. However, mechanical instability, such as shifts in the X-ray tube during *FDD* adjustments, was observed to influence beam consistency, though the deviation remained within acceptable thresholds. These outcomes highlight two primary contributors to field misalignment: the geometric divergence of the X-ray beam at short distances and minor instability in the generator system during mechanical repositioning.[52,53]

Beyond beam perpendicular and beam alignment, this study also examined object magnification. In this context, we analyzed the influence of object-to-detector distance and focus-to-detector distance (*FDD*) on the resulting magnification factor. The phantom

Table 6
The average displacement distance and shift of test points relative to the central point on the geometric phantom.

Layers	Original Distance (mm)	Object Distance from the Center (mm)			Object Displacement (mm)		
		FDD 70 cm	*FDD* 60 cm	*FDD* 50 cm	*FDD* 70 cm	*FDD* 60 cm	*FDD* 50 cm
Bottom	55.0	57.1 ± 0.1	58.7 ± 0.1	60.9 ± 0.1	2.1 ± 0.1	3.7 ± 0.1	5.9 ± 0.1
Middle	55.0	65.8 ± 0.1	66.8 ± 0.1	70.7 ± 0.1	8.6 ± 0.1	11.8 ± 0.1	15.7 ± 0.1
Top	55.0	75.0 ± 0.1	79.9 ± 0.1	85.4 ± 0.1	20.0 ± 0.1	24.9 ± 0.1	30.4 ± 0.1

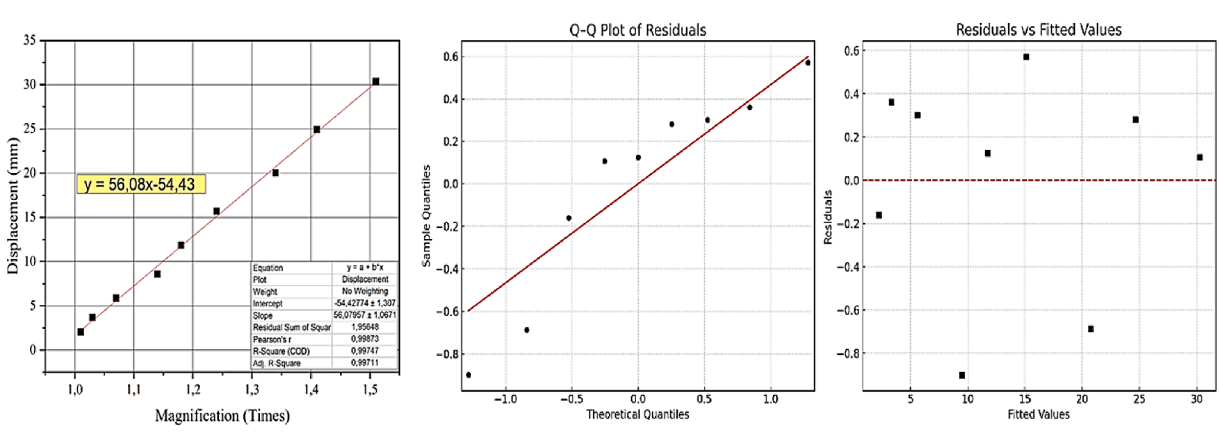

Figure 9. (a) Diagnostic plots for OLS regression analysis between magnification and test-point displacement. (b) Q–Q plot of the residuals shows that data points closely follow the reference line, suggesting normal distribution of residuals. (c) Plot of residuals versus fitted values shows no discernible pattern, indicating scedasticity and supporting the adequacy of the linear model assumptions.

layer levels represent the distance of the test point from the detector. The test object diameter significantly increases in the middle layers compared to the lower layers. The most significant increase in test point diameter occurs in the upper layers. This is because the position of the upper layers is considerably farther from the detector, at 210 mm.[54–56] On the other hand, the results demonstrate that as the *FDD* increases, the test object magnification decreases. Conversely, smaller *FDD* produce greater magnification.[56,57] The distance between the object detector and *FDD* is directly related to the object's magnification factor, as shown in the graphs in Fig. 7(a) and (b). Both graphs demonstrate that an *FDD* of 70 cm, with the test point positioned in the lower phantom layers, yields an acceptable magnification (0.85 %), remaining below 2 % of the test point's actual size. In contrast, at shorter *FDD* and greater phantom layer distances from the detector, the magnification percentage increases significantly, exceeding 2 % of the actual test point size. This result is consistent with the findings of previous studies, Olmedo-García et al. (2018), who identified FFD as the principal determinant of magnification, and Read et al. (2012), who reported an approximate 1 % magnification error for each centimeter increase in *ODD*.[26,28] Our results confirm that greater phantom-to-detector distance and shorter *FDD* substantially amplify magnification.

The significant increase in the magnification factor observed in this study is not solely attributed to *FDD* and the varying distances between phantom layers and the detector, but also to the relatively small diameter of the test point. Smaller objects are more susceptible to apparent size distortion due to minor shifts in geometric alignment or distance variation.[58] This sensitivity amplifies the effect of magnification, especially when the object is placed farther from the detector and under low *FDD* conditions.[11] These findings provide a valuable reference for radiodiagnostic applications, especially when estimating the relative magnification of larger anatomical structures. A better understanding of the magnification behavior of small objects under controlled conditions may enable practitioners to extrapolate and correct for magnification in clinical imaging, especially in scenarios where spatial constraints hinder ideal positioning.[11,26] Therefore, this magnification data can support the calibration of imaging systems and aid in achieving more accurate size estimations in diagnostic radiology. While these findings align with established magnification theory, the present study extends current knowledge by providing quantitative experimental evidence and visual validation of beam deviation, thereby reinforcing the practical significance of geometric calibration in digital radiography.

In the final stage of the analysis, we evaluated the displacement of test points to quantify the extent of spatial deviation relative to their expected geometric positions. This displacement reflects the degree of shift that occurs as a function of the distance between the object and the detector, a phenomenon well-documented in radiographic imaging, where an increased *ODD* results in magnification of the projected image.[59,60] As presented in Table 6, the test points embedded within the geometric phantom exhibited a measurable average deviation from the phantom's central axis, influenced by both the object-to-detector distance and the *FDD*. These findings are directly associated with the magnification experienced by the test points. Statistical analysis revealed a strong linear correlation between magnification and test point displacement, with a Pearson correlation coefficient of $r = 0.9987$ ($p < 0.0001$). This indicates that as magnification increases, driven by greater object-detector or object-source distance, there is a proportional increase in the spatial displacement observed in the radiographic image. Furthermore, linear regression analysis produced the following equation: $y = 56.08x - 54.43$. This model suggests that each 0.01-unit increase in magnification results in an approximate displacement increment of ± 0.56 mm, underscoring the sensitivity of positional accuracy to minor variations in geometric magnification. Diagnostic tests confirmed the robustness of the model. The residuals were normally distributed (Shapiro–Wilk test, $p = 0.1570$) and displayed random dispersion for the fitted values, indicating the absence of heteroskedasticity (Breusch–Pagan test, $p = 0.895$). These results validate the consistency of the magnification–displacement relationship across the dataset, unaffected by uncontrolled variance or outlier influence.

Nevertheless, this study has several limitations. First, it was conducted using only a single digital radiography system, which restricts the generalizability of the findings across different equipment types and manufacturers. Second, the investigation focused solely on variations in *FDD* and *ODD* under controlled experimental conditions, without accounting for other influential variables such as focal spot size, tissue thickness, or patient motion, all of which may affect image quality in clinical practice. In addition, phantom exposures were performed only once for each *FDD* setting, thereby not addressing the potential dynamics of X-ray beam misalignment under repeated use. Future research should consider the development of phantoms with more complex geometries and object sizes that more closely approximate human anatomy, alongside evaluations conducted on diverse radiographic systems with varying exposure parameters under clinically relevant conditions. Further studies should also investigate software-based or algorithmic approaches to magnification correction, enabling more direct translation of these findings into diagnostic radiology practice.

Conclusion

This study successfully introduced and validated a geometric phantom as a multifunctional tool for evaluating the geometric accuracy of digital radiography systems. The phantom enabled precise measurements of beam perpendicular, beam area alignment, magnification, and object displacement, which are highly influenced by object-detector distance and *FDD* in digital radiography. Shorter *FDD* increased angular deviation and field misalignment, though still within acceptable limits. Magnification increased significantly with greater *ODD*, particularly for small-diameter objects, and exhibited a strong linear relationship with object displacement (r = 0.9987). These findings support the importance of geometric calibration to maintain image accuracy and diagnostic reliability.

Ethics approval and consent to participate

Not Applicable.

Availability of data

Data required for this study may be made available by the author(s) upon reasonable request.

Author contributions

NLSM: Conceptualisation, Methodology, Investigation, Writing-Original Draft preparation.

IZTD: Data curation, Visualization, Investigation.

WMD: Data curation, Writing- Reviewing and Editing.

AFA: Validation, Visualization,

RR: Software.

GBS: Conceptualisation, Methodology, Supervision, Writing- Reviewing and Editing.

Generative AI use

During the preparation of this work, the author used *QuillBot* and *Grammarly* to enhance the language, including checking grammar, improving sentence structure, and refining the overall writing style. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Conflict of interest statement

None.

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